

Supplementary Material

ONCOPLEX: An Oncology-Inspired Hypergraph Model Integrating Diverse Biological Knowledge for Cancer Driver Gene Prediction

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1 Experimental setting

1.1 Hyperparameters optimization

A successful machine learning model depends on the selection of suitable hyperparameters, which makes careful tuning essential. We perform hyperparameter optimization using 5-fold cross-validation. The labeled data were divided into K folds, where (K-1) folds were used for training, and one fold was kept for testing. Within each training fold, an inner loop applied a simple grid search to explore combinations of hyperparameters and identify the optimal configuration based on validation performance. We tuned the following hyperparameters:

- Number of hidden units: [64, 128, 256]
- Number of layers: [2, 3, 4]
- Learning rate: [1e-3, 5e-4]
- Weight decay: [1e-3, 1e-4]
- Dropout: [0.25, 0.4, 0.5]
- Positive class weight: [0.2, 0.4, 0.45]

The **outer loop** was then used to evaluate the model with the best hyperparameter combination and report the average and standard deviation for the three metrics AUPRC, AUROC, and the F1 score. This evaluation provides a robust and unbiased estimate of the performance of the model. We repeat the same procedures for the specific cancer setting. In some types of cancer, the best set of hyperparameters is identical, while in other cases, a different set is optimal for that specific cancer.

1.2 Datasets

Driver gene labels: We collect driver gene sets for both pan-cancer and cancer-specific driver genes from the following resources:

- Network of Cancer Genes (NCG) v6.0¹: http://ncg.kcl.ac.uk/download_file.php?file=cancergenes_list.txt
- DigSee Database²: <http://210.107.182.61/digseeOld/>
- COSMIC Cancer Gene Census (CGC) v91 and COSMIC Mutations in Census Genes³: <https://cancer.sanger.ac.uk/cosmic/download>
- IntOGen tumor-specific driver lists v2024.09.20⁴ <https://www.intogen.org/download>

Gene sets used for biological validation: To further confirm our results, we assessed whether the newly predicted genes appeared in the following cancer gene databases:

- **OncoKB:** A manually curated resource of cancer genes annotated based on validated oncogenic effects. Only high-confidence cancer genes, supported by evidence from more than three independent sources, including clinical studies and association with FDA-approved drugs, were included in this set⁵: <https://www.oncokb.org/cancerGenes>
- **ONGene:** A collection of human oncogenes curated in the literature⁶.
http://ongene.bioinfo-minzhao.org/ongene_human.txt
- **CancerMine:** A text mining-based database of cancer genes extracted from the published literature⁷.

2 Results

2.1 Analysis of newly predicted genes by OncoPlex across 11 cancer types

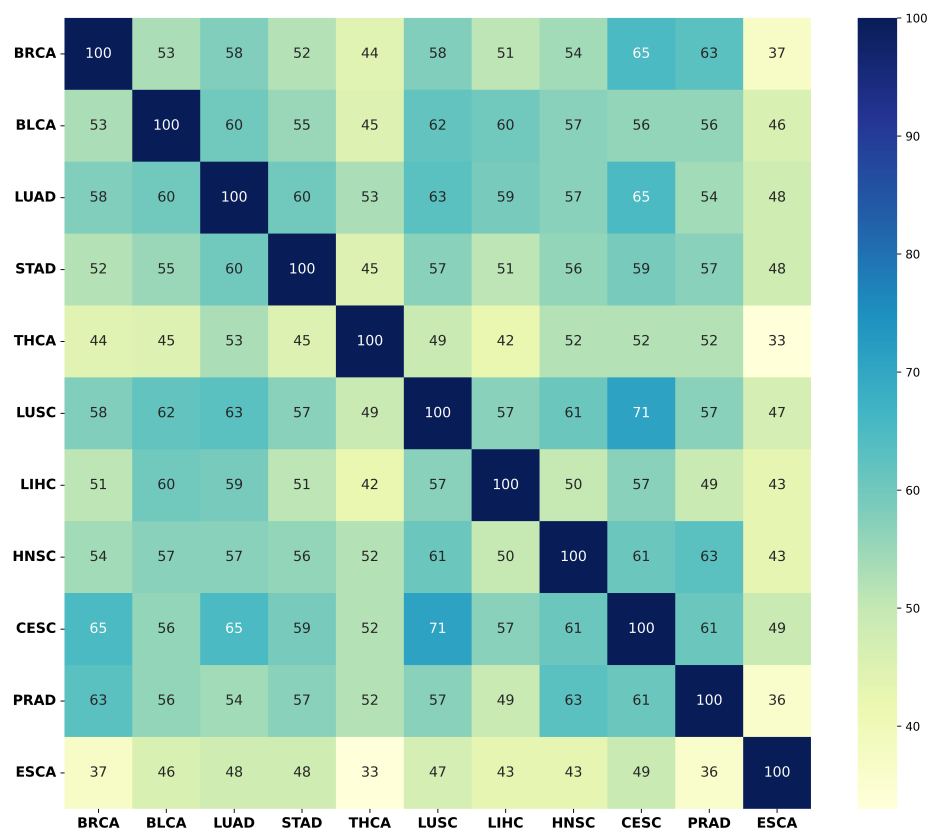


Figure 1. The overlap among the 100 newly predicted genes across different cancer types. For each cancer type, we evaluate how its predicted genes intersect with those of other cancers. As shown, some cancer types, such as BRCA, exhibit high levels of overlap with many others, suggesting shared molecular characteristics. In contrast, cancers like ESCA show minimal overlap with most other types, except specific cases such as CESC, STAD, and LUAD, where moderate gene-level similarities are observed.

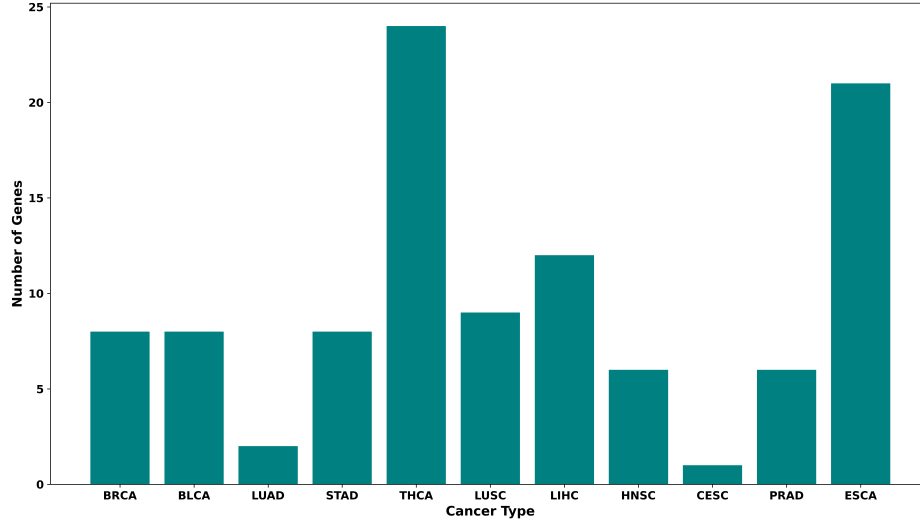


Figure 2. The number of unique predicted genes identified for each cancer type. We analyze the genes that are exclusively predicted for a specific cancer type. Among all cancers, THCA and ESCA exhibit the highest number of unique genes, which can be attributed to their limited overlap in pathways with other cancers. This reduced pathway overlap results in fewer shared genes with other cancer types.

3 Methods

3.1 Node features

We used three types of node features: incidence-based features, core omics features, and comprehensive features.

Incidence-based features: These features were derived solely from the hypergraph topology, without relying on any predefined biological node features. Each gene is represented by a binary matrix that captures its position within the pathway hypergraph. This setup ensures that the only source of information comes from the higher-order pathways connection.

Core omics features: These features were derived from multi-omics data collected in 16 types of cancer in the TCGA database⁸. For each type of cancer, we included three omics layers: Single-Nucleotide Variants (SNVs), gene expression, and DNA methylation. Following⁹, the preprocessing steps are described below.

- **Single nucleotide variants:** We extracted the Single nucleotide variant from the TCGA Mutation Annotation Format (MAF) files. We computed the mutation frequency mf for each gene by normalizing the number of non-silent mutations by gene length: $mf = \frac{\sum m}{L}$, where m is the number of mutations and L is the gene length. We excluded ultra-mutated samples to reduce noise due to genome instability.
- **RNA-Seq gene expression data:** TCGA provided RNA-Seq data as normalized FPKM counts. After batch correction, we computed the logarithmic₂ fold change between tumor and matched normal samples: $\log_2 FC_g = \log_2 \left(\frac{FPKM_g^{\text{tumor}}}{FPKM_g^{\text{normal}}} \right)$, where $FPKM_g^{\text{tumor}}$ and $FPKM_g^{\text{normal}}$ represent normalized expression of the gene g in tumor and normal samples, respectively.
- **DNA methylation:** We used beta values (ranging from 0 to 1) to compute differentially methylated regions. The logarithmic₂ fold change at the gene level was calculated as: $dmc_i = \frac{1}{|S_c|} \sum_{s \in S_c} \log_2 \left(\frac{\beta_i^t}{\beta_i^n} \right)$, where β_i^t and β_i^n are the beta values for the tumor and normal samples, respectively.

Each omics type generated a gene-by-sample matrix, and we combined the three matrices per cancer type to form a final multi-dimensional feature representation. In the pan-cancer setting, this resulted in a **48** dimensional feature vector per gene. For cancer-type-specific networks, we used only the corresponding omics features from that cancer, resulting in a **3** dimensional feature vector per gene.

Comprehensive features: The last set of node features, termed comprehensive features, is a 44-gene, epigenetic, and functional set of features collected from previous cancer driver prediction methods¹⁰. These features include biological properties at the gene level, such as histone modifications, evolutionary conservation, and gain/loss of function indicators. They are particularly useful in distinguishing oncogenes from tumor suppressor genes.

In the pan-cancer graph, we trained using all **44 comprehensive features**. In cancer-specific settings, we integrated these comprehensive features with the cancer-specific omics features described above, enabling the model to jointly learn both cancer-type-specific and shared cross-cancer properties. (See Supplementary Table 1).

Supplementary Tables

The following supplementary tables (Supplementary Tables S1–S12) are provided as a single Excel file located in the same folder as this document, with each table presented in a separate worksheet.

Supplementary Table S1: Comprehensive features

Detailed description of the comprehensive feature set used in the second experiment to distinguish tumor suppressor genes from oncogene drivers between cancer types. The 44 features include mutation-related characteristics (e.g., missense and loss-of-function mutations), epigenetic signals such as DNA methylation and histone modifications, and an additional phenotype-based feature designed to improve predictive performance.

Supplementary Table S2: Cancer-specific driver and passenger gene statistics

Statistics of driver genes for each type of cancer based on the COSMIC and DigSEE databases. Passenger genes are shared across all types of cancer in the pan-cancer setting. The reported gene counts correspond to the genes included after hypergraph construction, with all genes successfully mapped to the network.

Supplementary Table S3: Pan-cancer results

Complete performance results for ONCOPLEX and all baseline methods evaluated on the pan-cancer dataset. Reported metrics include the AUPRC, AUROC, and F1 score, each with the corresponding standard deviation.

Supplementary Table S4: Pan-cancer ranking results

Ranking-based evaluation results for ONCOPLEX and baseline methods in the pan-cancer setting. Metrics include Precision@k and cumulative hit counts (Hits@k) for values of k ranging from 1 to 50, reflecting the relatively large number of known pan-cancer driver genes.

Supplementary Table S5: Cancer-specific results

Performance results for ONCOPLEX and baseline methods evaluated separately for each cancer type. Evaluation metrics include the AUPRC, AUROC, and F1 score, reported as mean $\pm$ standard deviation.

Supplementary Table S6: Cancer-specific Precision@k results

Precision@k values for k = 1, 3, 5, and 10 for ONCOPLEX and baseline methods across individual cancer types. Smaller k values are reported due to the limited number of known driver genes for each cancer type compared to the pan-cancer setting.

Supplementary Table S7: Cancer-specific Hits@k results

Raw counts of correctly recovered true driver genes (Hits@k) at k = 1, 3, 5, and 10 in cancer-specific evaluations. These results complement the Precision@k values reported in Supplementary Table S6.

Supplementary Table S8: Newly predicted genes

Complete list of 30 newly predicted candidate genes for each cancer type. Genes are classified as either common drivers or novel candidates not previously reported as cancer driver genes, and supporting evidence from established cancer-related databases is provided where available.

Supplementary Table S9: Novel cancer genes

Top-ranked novel candidate genes identified in selected cancer types. For cancer types in which no novel genes appear among the top predictions, no entries are reported. The potential biological relevance of these genes was assessed through a comprehensive manual literature review.

Supplementary Table S10: KEGG pathway enrichment analysis for newly predicted genes

Results of KEGG pathway enrichment analysis conducted on newly predicted genes in BRCA, HNSC, and STAD. Only significantly enriched pathways are reported.

Supplementary Table S11: Hallmark gene set enrichment analysis for newly predicted genes

Significantly enriched Hallmark gene sets associated with newly predicted genes across cancer types, reported using the same statistical criteria as in Supplementary Table S10.

Supplementary Table S12: Performance of ONCOPLEX with and without CNA features

Comparison of ONCOPLEX performance, measured by AUPRC, using core omics features with and without the inclusion of copy number alteration (CNA) features. The results are presented for both pan-cancer and cancer-specific evaluation settings.

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